

Duccio Dogheria

INTERNET ARCHIVE



**an unofficial guide
to the World's most
innovative digital library**

le **STRADE BIANCHE**
di STAMPA ALTERNATIVA

“ **Non esistono
diritti d'autore,
solo doveri** ”

Jean-Luc Godard

MILLELIREPERSEMPRE

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What is Internet Archive

Internet Archive (www.archive.org) stands as the world's foremost digital library, liberated, free, devoid of advertisements, and driven by non-commercial objectives. Alongside Wikipedia, it ranks among the few non-profit entities listed among the 300 most visited sites globally. Established in 1996, its primary aim was to offer unrestricted and universal access to knowledge by making various types of documentation available online: primarily books and magazines, but also encompassing images, videos, audios of all kinds, software, and even web pages. Currently (November 2024), Internet Archive hosts over 42 million texts, 13 million audio files, 13 million videos, nearly 3 million TV broadcasts, almost 5 million images, over a million software files, and 260 thousand audio concerts, all organized within 2,5 million collections.

The founder, Brewster Kahle (1960), holds a significant position in the history of computing and the digital era. Following his graduation from the Massachusetts Institute of Technology in 1982, he served as chief engineer for Thinking Machine Company, and in 1992, he was among the developers of the Internet WAIS search system, a precursor to the World Wide Web. In the same year he founded Internet Archive, Kahle also co-founded Alexa Internet, which was later acquired

by Amazon. Thanks to its inherently decentralized vision of the web, aiming to make every user an active participant, Internet Archive marvelously hosts an array of content, from medieval parchment manuscripts to those written on palm leaves from Southeast Asia, from incunabula to native digital documents, from vintage computer viruses enjoyed safely through simulators to hundreds of thousands of digitized 78 RPM records, from underground publishing to documentary sources of all kinds (audio, video, web) for contemporary historians, from sci-fi B movies to 3D printer programs (as early as the first months of 2020, there were STL files available for printing COVID-19 masks at home), from children's literature to academic literature spanning every research field.

Internet Archive (hereafter IA) features accounts from major universities, museums, libraries, and other international institutions, but also and especially pages created by individual scholars, collectors, and enthusiasts eager to share their interests with others by offering their collections. In its own way, IA is also social! All uploaded documents can be shared and reviewed, with comments often supplementing and enriching their descriptions in a participatory manner.

The extensive digitization effort promoted by IA and its users also serves as an excellent response to the

longstanding problem of media obsolescence , including digital media, which is often overlooked: Digital decay is constantly looming, as media such as floppy disks, CDs, CD-ROMs, USB drives, memory cards, and external hard drives often have a shorter and more fragile lifespan compared to older media such as vinyl records, audio cassettes, and film reels.

Everything published on IA can be freely accessed online through the browser and mostly downloaded, although it's rarely necessary since the documentation is always accessible – connection permitting – and secure.

Reliability is ensured not only by nearly 30 years of experience but also by the powerful servers located at IA's headquarters in San Francisco (housed in a former neoclassical-style church in the Richmond district, chosen for its façade resembling that of a classical temple, the logo of IA) and mirrored copies in the Bibliotheca Alexandrina in Alexandria, Egypt, and in Amsterdam. Additionally, in 2022, Internet Archive Canada (<https://internetarchivecanada.org>) was officially inaugurated, a twin non-profit entity – though legally autonomous – of Internet Archive.

Using IA for research is simple, easy, and intuitive: from the homepage, you can utilize the free-text search box, which captures not only author and title but

also all metadata associated with the individual document; or, once you become more proficient, you can undertake an advanced search; or yet, especially recommended for initial explorations of the platform, you can browse through various sections of the site, found in both the top bar and the center of the screen, identified by pictograms. Particular attention to content accessibility has been directed towards people with visual impairments, for example, by making many texts available in DAISY (Digital Accessible Information System) format.

Exploring IA's content is so straightforward and intuitive that it seems unnecessary to spend more words on it; just let your interests and curiosity guide you to discover the hidden worlds within this portal, including special sections like *Open Library and Books to Borrow*, which allow account holders controlled digital lending (CDL) of recently published books protected by copyright, for a specified but renewable period.

We will dedicate only a brief chapter to search methods in conclusion, because the intention of these pages is rather different: to introduce an active use of IA, allowing anyone to take the first steps in creating an account, populating a content page, and enriching the metadata to subsequently optimize searches by users. We will explore how to enhance the presentati-

on of cards and provide some small, valuable tips to address some of the most common doubts among newcomers, which may not be addressed in available tutorials but only in the experience of those who have been operating on the platform for some time. For anything not found here or in the tutorials, we encourage you to consult the Help Center pages (<https://help.archive.org/>) or the recent New to the Archive? section, or directly contact the Internet Archive staff (info@archive.org), who have always been responsive and punctual in their responses.

Creating an account

From public cultural institutions to individual enthusiasts, anyone can freely and easily open and manage an account on IA. Doing so is straightforward: from the portal's homepage, click on *Sign up* to the right of the top bar, and an entry form will appear (fig. 1).

You'll need to provide a valid email address, to which a confirmation email will be sent; the name you want to give to your user page, visible to other users; a password for access, which will then be automatically stored for subsequent logins.

All entered data can be easily modified as needed, except for the address of your user page (which can still be changed by contacting IA staff).

Once the user page is open, although it may still lack content, it can be enriched by a description of any length, visible beneath your name; to insert it, simply click on Edit at the top right and then fill in the Description field. You can also add a logo image to your page, which can also be modified at any time: just upload an image or drag it to the icon to the left of your name.



Sign up for an account

Get your virtual library card and access 65+ million free and borrowable books, movies, audio, images, software, and more! Already have one? [Log In!](#)

 Accedi con Google

_____ Or _____

Enter email address Up to 127 characters

Choose a screen name (this is how others will see you) 3 to 127 characters

Choose a password 3 to 100 characters



Fig. 1 - What you need to create an account

Uploading an item

After creating your user page, the next step is populating it with content. Uploading a document is straightforward: first, click on Upload in the top right corner. Unless you are a musical band – in which case you’ll need to select the Live Music Archive uploader option –, in the dialog box that appears, click on Upload files and then select or drag the file (of any type) you want to upload. Unlike many other platforms, IA imposes no limits on file size or quantity. For system convenience, IA generally suggests not exceeding 50 GB per document and, in the case of a document consisting of multiple files (such as a series of photographs), not exceeding a total of 1000 files.

In the dialog that opens, you’ll need to input all metadata (data about data) related to your document (hereafter referred to as “item”: this is the term used by the platform). Metadata is essential not only for proper archiving of your document but also to ensure that your item doesn’t remain an isolated star in the IA galaxy but is easily searchable by other users. IA adheres to the international metadata standard called Dublin Core: some fields, marked with an asterisk, are mandatory and will already appear pre-filled during input (but still editable). IA guarantees maximum freedom in filling out and managing metadata: except for

the marked fields, there are no insertion constraints, no imposed descriptive standards, only simple suggestions, which you can find in the Help section (Example of good metadata for items).

This freedom also extends to the ability to add additional metadata to your items, both to describe any specificities of the uploaded resource and to enhance the user search experience: do you want to add a specific metadata about the signature/location of the item in your library or collection? You can do that. Do you want to add metadata about the printing technique of your item (e.g., for photographs or graphic works)? You can do that. Do you want to add metadata linking to the Wikidata page of the author of the work? You can do that...

Too much freedom can sometimes be exhilarating, and the risk of confusion is always around the corner. Therefore, it's essential to have at least some coherence and uniformity in metadata entry within the same collection: for example, if you decide to enclose the titles of periodicals in quotation marks, ALWAYS write them in quotation marks; if you use a specific format for the Creator field (i.e., the author), such as name-surname or surname-name separated by a comma, ALWAYS use that format consistently. Therefore, we advise you to draft your own rules for metadata entry, referring as needed to the best practices found on IA and always

keeping in mind what your goals and priorities are. Now let's briefly summarize the fields you'll be required to fill in on the input screen (Fig. 2):

Page Title is the first mandatory field: it will be pre-filled with the same name as the uploaded file. You can modify it as you like, but, as mentioned, consistently with other inserted items: for example, if you use the string Author's name and surname, Title, Year of pub-

Click on any field below to edit it		
Page Title *	Prova Caricamento Internet Archive. 7z	✓
Page URL *	https://archive.org/details/prova-caricamento-internet-archive.-7z	✓
Description *	Add a description of the item page	✓
Subject Tags *	Add keywords, separated by commas	✓
Creator	Creator of the content	✓
Date	Date work was created/published	✓
Collection *	Community Audio	✓
Test Item	No	✓
Language	Language of the work	✓
License	No license selected	✓
More Options	Add additional metadata...	

Fig. 2 - The metadata entry mask

lication (e.g., G. Debord, The Society of the Spectacle, 1967) in the Page Title field, use that formula consistently. From the Page Title field, the Identifier will be automatically derived, which is the unique alphanumeric reference to your item within IA.

Page URL is the web address of your item, always starting with <https://archive.org/details>, followed by a specific part about the item (e.g., <https://archive.org/details/d.-dogheria-internet-archive.-guida-non-ufficiale-alla-piu-innovativa-biblioteca>). For convenience – but feel free to approach this with maximum freedom – it can be useful to copy and paste the content of the previous Page Title field, i.e., the title of your item, so that the corresponding URL can immediately identify the content.

Description is a substantially free field and can contain very lengthy descriptions; you can format the text and insert links for further information. Each term entered in this field is indexed and therefore searchable by the search mask.

Unless you want to create specific metadata, you can insert, for example, signatures of printed works, any ownership notes (e.g., ownership signatures or bookplates of previous owners), etc., here. If you have descriptions from your databases, of course, the simplest thing is to copy and paste them.

Subject Tags are very useful metadata for searching: tags are used to create semantic relationships among various items uploaded to Internet Archive, both in general – to facilitate the work of researchers interested in a specific topic – and within the collection, to group related documents linked by the same theme or occasion. Consider translating at least some of the tags into English, given the international audience of the platform. During insertion, simply separate tags with a comma; you can then modify them (delete, correct, add, etc.) at any time using the Edit function, but it's a bit more complicated; therefore, we recommend spending some time on this, perhaps creating simple models in Word if you need to insert multiple related items, then proceed with copy and paste and modify as needed. Around ten not overly generic tags are ideal (fig. 3).

Creator is the author of the document: the platform allows you to input only one author during insertion, but if there are multiple authors, others can be added via the More options. Add additional metadata function: you'll specify the field you want to add (Creator) on the left and the value (the name and surname of the second author, according to the format you've chosen) on the right. Alternatively, you can add other authors later via the Edit function. Keep in mind that any secondary authors (e.g., cover graphic designer,

Topics	American novel, fiction, F. Scott Fitzgerald, Jay Gatsby, West Egg, East Egg, Daisy Buchanan, Tom Buchanan, Jordan Baker, Nick Carraway, American dream, Class permanence
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Fig. 3 - Tags (Topics) associated with an edition of *The Great Gatsby* by F. Scott Fitzgerald.

translator, etc.) can be mentioned in the Description field and can still be searched on IA, as they are always indexed.

Date is the date of the item, expressed in numbers according to the YYYY/MM/DD (year/month/day) format. It's straightforward for precise dates, especially for editions (books, magazines, newspapers), but what if the date is open or uncertain? Since the insertion dialog only allows you to indicate a precise numerical value, our advice is to leave the field empty, as it's not mandatory, and to intervene later (we'll see how), via the Edit function.

Collection is the collection in which to place the item: if you have already created one after reaching a minimum of 50 items (we'll see shortly how to do this), simply enter the first letters of the collection's name,

and after a few seconds, the platform will suggest the Collection, to be confirmed with a click; if you haven't created one yet (or don't want to create one), IA will automatically save the item in a general collection, depending on the format of the uploaded file (e.g., Community Audio for audio tracks, Community Video for videos, etc.). A word of caution: if for any reason you prefer to prevent the download of the uploaded item while still making it fully viewable and searchable, you'll need to indicate the value `stream_only` in the Collection field. Moreover, the reliability of IA has partially rendered the choice to download items obsolete, as they are always and conveniently available online, moreover on a completely mobile-friendly platform.

Test Item if selected, allows you to upload an item for only 30 days, after which it will be automatically deleted. Unless it may be useful for a particular occasion, we recommend leaving the field as it is at the time of upload, i.e., pre-filled by default to no.

Language is the language of the item, obviously not always present, as in the case of images. If the item has multiple languages, indicate the primary one during insertion, or one of the primary ones: you can then add the other languages using the Edit function. In addition to being able to search for documents in

a particular language, correctly entering the language field enhances the OCR with which the printed document is processed, also enabling the Read this book aloud function (symbol of a pair of headphones: you can find it in the document's function bar), which allows you to listen to the text being read, although the quality is not that of an audiobook.

License is the type of usage license to which you bind the document. Specify the most appropriate one (see the types of Creative Commons licenses at <https://creativecommons.org/share-your-work/cclicenses>); the field, not being mandatory, can be left blank, as it appears by default (Leave license blank).

More Options is a repeatable field that allows you to add additional metadata: for example, you can add a second Collection here (such as `stream_only`), or a second Creator, specifying the field on the left (key) and the value on the right, or add additional metadata, as you please.

Once this simple form is filled out, a click on Upload and create your item will start the upload process, the duration of which will depend on the file type and size. Once the item is uploaded, you will initially see the message There is no preview available for this item. Don't worry: if you scroll down, you will first see

the card you filled out with all the metadata and another notice on a blue background indicating that the item is still being processed: your item, before being visible, undergoes a very complex series of transformations and conversions into other formats, which subsequently allows you to take advantage of all the platform's features, such as automatic OCR for all texts. Before being correctly displayed, and able to modify its metadata if necessary, some more time will pass, even over an hour, but the final result will reward your patience!

If what has been described seems complicated to you – but I assure you it isn't: it just requires a bit of practice – there is a video tutorial on Internet Archive by Alexis Rossi, *Uploading Overview* (<https://archive.org/details/help-upload-overview>), which explains the essentials in a few, very simple steps in just 5 minutes.

Books, magazines & co.

As a digital library, it's natural that the most prevalent and consulted item type on Internet Archive is related to printed texts: books, magazines, and more. Textual content is best enjoyed on IA when uploaded in PDF format. While most professional scanners allow you to digitize a work directly into PDF format, if, for any

reason – including the need for post-production of images – you cannot or do not want to save your digitized text directly in PDF, you can do so later without any issues. The important thing is that during the digitization process with a scanner (or camera), you produce reproductions of individual pages in the correct order, from the cover to the back cover, and that the sequence of pages corresponds to a relative sequence in the filenames. After any post-production phase (straightening, image clipping, etc.), you need to convert the individual reordered images into a PDF file. There are many tools for doing this, including free and online ones. If you have Photoshop, after opening the program, select File – Automate – PDF Presentation; then select all the JPG images from the Browse button, click Open, then Save, and finally OK: the PDF file will be created in a few moments, ready to be uploaded to IA.

Creating a *Collection*

Having learned how to correctly upload items to the platform, you are ready to bring your Collection to life, or rather, your digital library. To do this, you'll need to have uploaded at least 50 items to IA. Your user page essentially serves to manage the entries and changes to your digital library, even though there are quite a few users who, for various reasons, limit themselves

to creating only one user page despite having surpassed the 50-item mark: the contents are still visible and searchable, although dispersed among various community collections.

The procedure for creating a Collection is straightforward. Once you have reached the minimum threshold of 50 items, you'll need to send an email to IA (info@archive.org) specifying:

- The email address of the IA account that uploaded the items to be part of your digital library (Collection).
- The list of all the URLs of your items, or – which is much simpler – a tag (or combination of tags) present in all your items, and only in your items. For example, a tag that uniquely identifies your Collection (first, check the metadata search form to ensure it hasn't already been used by others).
- The name you want to give to your Collection and a brief presentation of it. The presentation, which you can write in any language, can be modified and supplemented at any time, even with the addition of images.
- An Identifier, a name identifying your digital library, which will become part of the URL of your collection.

Neither spaces (replaceable with hyphens if it consists of multiple words) nor special characters (e.g., Gotham-City-Library) are allowed.

- At least one *Subject Tag* identifying the Collection (e.g., the name of your Collection).

- The email address or addresses of the IA accounts authorized to operate within the Collection. You can naturally add or remove page administrators at any time, remembering that there are no levels of authority.

- A logo of your digital library (optional but recommended).

- The order in which you want your items to appear by default in your digital library, such as by date or by author (optional but recommended).

This order can be changed by you at any time, both in default mode and temporarily, using the sort by function, which allows you to change the presentation order according to multiple parameters (document date, document insertion date, document review date, alphabetical order by author, alphabetical order by title, views).

After receiving the email, the IA staff will create your

digital library in a few days, subsequently sending you a confirmation email.

Over time, as your digital library becomes populated with content, you may decide to create sub-collections, grouping parts of your collection by type. For example, if you have created the digital library of a municipal library, one *sub-Collection* could be dedicated to local history. The criteria and methods for creating a *sub-Collection* are the same as those seen for creating the *Collection*.

To give a couple of examples: the Getty Research Institute's digital library (<https://archive.org/details/getty>) has dedicated a sub-collection to the collection of Photographic Incunabula, the very first photographic books (1840-1870), and a second sub-collection to alchemy books; the Lazzerini Library in Prato, Italy, has instead dedicated two sub-collections to local periodicals ("L'Avvenire di Prato" and "La Fiamma") and two sub-collections to archival-library funds of individuals (Aldo Petri archive and Tullio Ceri archive).

Both the Collection (i.e., your digital library) and the sub-Collections (i.e., any sub-collections) offer great potential, from the possibility of an overall presentation of your digital collections to the ability to search (by metadata or textual) within them. Additionally, by

selecting About just below the collection presentation text, you can monitor the number of views by users from the opening date of your digital library, but also year by year or month by month. The views of the last 30 days are also geolocated, allowing you to discover the (often international) origin of your audience.

Post-production: Adding and editing metadata (and files) via the *Edit* command

We have emphasized the importance of metadata several times, both for archiving purposes and to expand the search possibilities for your item by other users. IA allows you to integrate, delete, modify, or add metadata to your item at any time. The operation is very simple: once you have opened your item, click on Edit (symbol: a pen) from the left menu, found just below the item title. You'll then have two options: click on I want to change the information (metadata) about my item (symbol: a painter's palette); we'll cover the other option (I want to change the files in my item) later. This will open the Metadata editor of your item, where you can freely add metadata, or replace or delete existing ones. Take some time to explore the complex metadata screen: each item is always accompanied by a brief explanation of its main characteristics. At the bottom of the list, you can add additional fields (and their values) by clicking on Add another

field. With this function, you can, for example, add a second language, a second author, or a second Collection to your item, if you haven't already done so during insertion. From the Metadata editor, you can also "force" the date field since, during insertion, it can only be filled with a numerical value: in all cases of open or doubtful dates, it will be possible to enter a date according to an absolutely free formula (ca. 1915, 1912-1915, ante 1915, 1820s, 15th century, 192?, etc.). The value entered will also be searchable, so remember to always use your own guidelines to ensure internal descriptive coherence on the page.

The Metadata editor is particularly valuable for the periodic checks we recommend you perform among the tags used for items in your Collection, through the faceted navigation menu found on the left (fig. 4). Checking them in their entirety, you may find spelling errors or notice similar tags that you can standardize in one way or another. In this case, by opening the Metadata editor, you can directly intervene on the tags, correcting, replacing, or deleting them.

From the Metadata editor, you can also change the display mode of your texts, which, for obvious reasons, is by default in double-page mode (Two-page view). Suppose, however, that the PDF you uploaded is a collection of press clippings or images that also include

textual content that you want to make searchable with full-text search. For purely aesthetic reasons, i.e., the graphic presentation of your document, you can decide in these cases on a default display of your item in single-page mode (One-page view) or thumbnail mode (Thumbnail view). To make this change, from the Metadata editor, via the More options function we have already encountered, add the bookreader-defaults field and the mode/1up value (for single-page mode), or mode/thumb (for thumbnail mode).

IA keeps track of every modification, no matter how minimal, of your item, from the moment of its upload. This function is of interest, also considering that each Collection can have multiple administrators, all of equal level and therefore all authorized to insert, delete, modify items, and their metadata. To reconstruct the history of these changes when necessary, just open the item and, in the URL, replace the term details with history.

Now let's look at the second option in the Edit menu, I want to change the files in my item (symbol: a butterfly). This function comes in handy on several occasions: for example, to add a cover image to an audio document, or again to add (or remove) one or more images to an existing gallery, or even to replace the PDF of an edition uploaded for which you have notic-

ed some errors in the page sequence.

To add a file, select I want to change the files in my item and then Add a file; select or drag the file in question, and finally click on Add files to existing item To delete a file (e.g., an image from a gallery), after clicking on I want to change the files in my item from the file tree contained in the item (Tree Root), select the file you want to delete, then right-click, click on Delete and confirm the deletion. Finally, to replace a file (e.g., a wrong PDF with a correct one), from I want to change the files in my item delete the wrong file with the procedure we just saw and replace it with the correct one, using the Add a file function seen earlier, with the caution that the correct file has the exact same name as the deleted one, otherwise part of the metadata will be lost.

Topics & Subjects



- ☐ Indians of North America 725
- ☐ Dakota Indians 345
- ☐ Smithsonian Institution 329
- ☐ Postage stamps 322
- ☐ Art 252
- ☐ Stamp collecting 238

[More](#) ►

Fig. 4 - The Faceted Navigation Menu of the Smithsonian Libraries and Archives – Digital Library

An Item, Multiple Files

An item can be associated with multiple files. As mentioned earlier, IA does not impose limits on the number of files, but it's suggested not to exceed 1000 files per item to avoid potential upload issues, given the lengthy upload times. IA supports hundreds of file formats across various media types, so encountering problems due to file type is extremely rare. The ability to associate multiple files with an item is particularly useful in various contexts. Let's illustrate the four cases that are particularly common:

Image Gallery: IA hosts over 5 million images, ranging from personal photo galleries (similar to Flickr) to reproductions of artworks from major international museums like the Metropolitan Museum of Art, to space images from NASA. Images can be uploaded and described individually or grouped together as an item in the form of a gallery. To create a gallery, you'll first need to prepare your images, naming them in a numerical sequence corresponding to their intended order. Consider whether to include a list of gallery images with their respective captions in the Description field during upload, and later add hyperlinks to individual images via the Edit function. For managing and organizing image items, draw inspiration from IA's best practices. Through the Edit function, you can add or delete images from your gallery at any time.

Audio: Even if an item consists of a single audio file, it's aesthetically pleasing to include a second file during upload, namely a cover image. This image could be a photograph of the original media (e.g., a record, cassette tape) or an image related to the content of the audio track (e.g., a portrait of the author or the occasion of the recording). It's more common for items to be associated with multiple audio files, such as individual tracks from a digitized 78 RPM record or individual segments from a conference. For uploading, simply select the group of files previously orde

red with a sequential number and, if necessary, a brief caption (e.g., the track title, optionally preceded by the author), to be used as the file name. Once the files are uploaded, a playlist with all tracks and their durations will be automatically displayed. IA handles large quantities of audio files from any device efficiently. Test how mobile-friendly IA is by searching for items like Gunsmoke single episodes directly from your smartphone on the platform (or find it on Google): you can quickly select one of the over 480 episodes of this western radio show set in the 1870s, which aired on CBS from 1952 to 1961.

Video: Similar to audio files, multiple video files can be linked to a single item, following the same process as for audio files. Having multiple video files collected in a single item offers numerous advantages, such as documenting individual presentations at a conference, serial videos, or assembling a series of videos related by theme or occasion, depending on your objectives.

Publications: In most cases, text items are associated with single files. However, an exception is possible for periodicals, which offer essentially three different upload options:

- Upload each individual issue of the periodical as separate items. For example, if you digitized a year's

worth of a monthly magazine, you would have a total of 12 items, each corresponding to a single issue, digitized into a single PDF file.

- Upload all individual issues of the periodical as a single item, linking all 12 PDF files to the item. In this case, the upload process is similar to that seen previously for items with multiple files: after uploading, an index of the uploaded issues, each individually selectable and viewable, will automatically appear in the item. The sequence and content of this index will be determined by the names of the individual PDF files.

- Upload an entire year of the periodical as a single item, using a single PDF file (as if it were a book). This is usually a practical and quick choice, useful for, for example, bound volumes of a periodical.

Searching on Internet Archive

Searching on IA is straightforward and intuitive. We'll provide some tips to enhance your experience. You can freely explore the platform, selecting the genre that interests you the most (Web, Texts, Videos, Audio, TV, Software, Images, Concerts, Collections), and then delve deeper into the collections and sub-collections that are presented to you. This is the easiest way to sample the variety of content available on IA.

For actual searches, simply click on the search bar (Search), and various options for simple search will appear. By default, the search is set to search metadata (Search metadata) – author, title, year, etc. – but for specific searches on multimedia items, especially useful for those dealing with contemporary subjects, consider other options such as Search TV news captions or Search radio transcripts. Full-text search (Search text contents) is extremely useful, allowing you to search for a single term or combinations of terms within one or more texts – whether across all texts available on IA (using the general search bar), within a specific Collection (from the search mask within that particular collection), or within a single text item, using the search mask on that specific item’s page (symbol: a magnifying glass).

To refine your free-text search, considering the vast digital heritage available on IA, you can use classic boolean operators, similar to those used on Google, also supported by IA. For example, if you’re looking for an item with both the metadata terms Marinetti and Fascism you would enter Marinetti AND Fascism in the search mask. If you’re searching for items that contain the term Marinetti but not Fascism the search string would be Marinetti AND NOT Fascism. Alternatively, if you want to find items containing either the term Marinetti or Futurism the search query would be Ma

rinetti OR Futurism.

In addition to AND, AND NOT, and OR, to search for an exact combination of terms, whether two words or an entire phrase, the exact search terms must be enclosed in quotation marks, for both metadata and full-text search. For example, to search for the opening line of Dante's *Inferno*, you would conduct a search in Search text contents mode by entering "Nel mezzo del cammin di nostra vita" in the query. Without using quotation marks, you would get more results, but without the precision you might desire for your search. For example, if you enter "the black cat sleeps" only those results with that exact sequence of words within quotation marks will appear, such as the black cat sleeps on the table; without quotation marks, you'll find other occurrences where the words are the same but in a different order, such as the black cat sleeps on the black sofa.

The ability for textual search allows you to explore a book (or potentially all books on IA, i.e., tens of millions of texts!) in few seconds. Test it out yourself, for example, by searching for the name of the protagonist, Dorothy, in the famous work *The Wonderful Wizard of Oz* by Frank Baum, using the Search icon.

To further enhance your search, you can use parentheses, similar to a mathematical operation. For in-

stance, Futurism AND (Marinetti OR Russolo OR Carrà) is the search key for items containing the term Futurism and, indifferently, Marinetti Russolo or Carrà.

Given the vast amount of multimedia documentation on IA, you can search for a term (or combination of terms) with boolean operators we've just discussed, combined with a specific media type (mediatype). Suppose you want to find an item containing the term Arthur Rimbaud that isn't a text item, as one might expect; in that case, the search query would be "Arthur Rimbaud" AND NOT mediatype:texts. Or, if you want to search for items related to the futurist musician Luigi Russolo that are only audio files, you would write "Luigi Russolo" AND mediatype:audio. In both cases, we've used the author's name in quotation marks to search for the exact name.

Regarding search, to learn more about IA searches using boolean operators associated with metadata, we recommend watching a video (Building Complex Search Queries) by Alexis Rossi, one of IA's pillars: (<https://archive.org/details/building-complex-search-queries>).

An extraordinary tool of Internet Archive is the Wayback Machine (<https://web.archive.org>), essentially a digital archive of the web that allows users to

browse through millions of websites, including those that have long since closed. The Wayback Machine was introduced in 2001 by Brewster Kahle and Bruce Gilliat, after more than 10 billion web pages were saved over five years, starting in 1996. Currently, there are over 866 billion web pages saved over time and accessible on the platform, totaling over 70 petabytes of data.

The potential of this tool is enormous, thanks in part to the option for users to autonomously save any website or web page, including social media posts, ensuring the ability to “eternally” preserve them with a permalink, a stable link over time. You can do this directly from IA by going to the media menu and selecting Web, then pasting the URL of the page you want to save in Save page now. To give just one example of the utility of this tool, with the onset of the Russia-Ukraine conflict, IA and its Wayback Machine initiated a campaign to archive Ukrainian websites (governmental, informational, and cultural sites) through the SUCHO project (Saving Ukrainian Cultural Heritage Online), given the concrete risk of server and data center destruction, and thus the disappearance of online content.

Work in Progress!

The aim of these few pages has been to introduce you to an active use of IA, entrusting your curiosity to explore the many other potentials of this extraordinary platform. In closing, we can only reiterate that IA is constantly evolving, with small and significant improvements, modifications, and bug fixes occurring daily. To visually understand this, simply use the Wayback Machine: try it out yourself, and you'll quickly realize the constant innovations, both graphical and content-related, introduced over the years by IA.

Among the most interesting developments that IA is working on is the adoption of the IIIF (International Image Interoperability Framework) standard for its viewer, a must for the most innovative digital libraries. IIIF is based on applications that allow image description, structuring of digital repositories, and search functionality, enabling advanced and interactive search capabilities for users, such as the ability to use and compare images from the web.

With IIIF, it's possible, for example, to virtually reconstruct dismembered manuscripts preserved in multiple institutions. If you want to learn more about IIIF developments in relation to Internet Archive, you can find more information on the platform's blog, [Making IIIF Official at the Internet Archive](#)

(<https://blog.archive.org/2023/09/18/making-iiif-official-at-the-internet-archive>). As we've mentioned, everything is rapidly evolving on IA, so expect surprises even regarding this standard.

For those fortunate enough to know a computer programmer, IA provides a Python library that allows for customization and enhancement of the platform's capabilities: almost all of IA's functionalities can be used and scripted from the command line or through Python scripts (<https://archive.org/services/docs/api/internetarchive>); furthermore, IA provides a set of APIs to interact with their services (<https://archive.org/services/docs/api>).

In conclusion, Internet Archive is a universal archive of knowledge in progress for almost 30 years, to which everyone can contribute, whether financially through occasional donations (because servers, technical personnel, and conservation projects naturally incur costs), or primarily by enriching it with content. Internet Archive, along with a few other initiatives like Wikipedia, has shown that another internet is possible, and to make this clearer to everyone, it has proposed a journey into a dark digital future where all knowledge is paid for, copyright is extended by another 200 years, content restrictions vary according to the personal data provided, and limitations are based on the

political area of reference. This journey, despite its premise, is quite entertaining and is called the Wayforward Machine, <https://wayforward.archive.org>, the alter-ego of the Wayback Machine: we invite you to explore it by inserting a link of your choice, discovering how bleak the future of access to knowledge could be, and what can be done to prevent it.

APPENDIX

Internet Archive and software emulators

By Emiliano Russo – Museo Interattivo di Archeologia Informatica (MIAI) / Museo dell'Informatica Funzionante (MusIF) – <https://miai.musif.eu>

In 2014, the Internet Archive launched The Internet Arcade project, a huge collection of video games produced in the golden age of arcades, between the late 1970s and the mid-1990s. The titles were made available by IA in streaming version, that is, playable directly on the web pages related to each published object. The initiative, welcomed by the enthusiasm of old and new gamers, was only the first of a series of thematic collections dedicated to the rediscovery of games and more generally historical software: it was followed by other “Software libraries” with thousands of programs created for hardware/software platforms now extinct.

From Pac-Man to the first versions of Word, these works of human ingenuity present peculiar problems from an archival point of view: they are “dynamic” objects - that is, they require user interaction to be truly enjoyed and understood - and were designed for a combination of hardware devices and software tools that have not been in production for a long time. A copy of the original executable of Doom, the seminal

video game from Id Software, cannot “run” like any other program on your computer, because your computer works in a way too different from that of a PC from 1993.

To overcome this inconvenience, the established practice in computer science is to resort to the use of an emulator, a particular program that “pretends” to be a hardware and/or software system different from the one on which it is executed, creating a virtual environment in which the programs written for the emulated platform can be executed unchanged.

With an emulator, for example, it is possible to run the original version of Super Mario Bros in a window of our desktop: the emulator will take care of “trick” the video game by presenting it with all the resources it expects to find within the system for which it was designed, in this case the NES, one of the very first game consoles produced by Nintendo. The input peripherals (original controllers/joysticks) will be combined by the emulator with those actually available on the PC (latest generation keyboard/controllers).

IA has integrated emulation services into its web portal, which are therefore immediately available to the end user: by consulting the page of an item that uses these services, an interactive box will be found in the preview area, activating which will allow you to run

the software and interact with it.

For example, at the URL <https://archive.org/details/ra-dio> it is possible to consult/use the original version of Ra-Dio by Lorenzo Miglioli, the first example of a hypertext book published in Italy in 1993, in its paper part (the scans of the book pages) and in its interactive part (a program distributed on floppy disk).

How does the activation of AI emulation services work?

In the following lines we will try to provide a small introductory guide to publish a “software object” from the past.

Let's say you have an old electronic publication in your hands - a fanzine for example - that in the era before the spread of the Internet was typically distributed on floppy disks. The first step in this computer archaeology operation consists, obviously, in transferring the contents of the diskette to a modern PC.

This operation can be complicated by factors such as the format (physical and “logical”) of the medium and its state of conservation. The era of the diskette was marked by a proliferation of incompatible formats; furthermore, the poor reliability of this type of sup-

port has always been well known, and the passage of time has only made the situation worse.

However, these problems are beyond the scope of this guide, and would deserve a long and in-depth technical discussion. So, here we will assume that we are in the most fortunate situation: that we are in possession of a 3.5-inch diskette, intact, functioning and initialized to the PC / MS-DOS format, the most widespread standard at the time of the personal computer boom, and still supported by the main operating systems in circulation.

To copy data from this particular medium you can use a USB floppy drive, a device that is no longer widespread but is still in production, easily available for a few euros. Most operating systems automatically recognize the drive and allow you to use it like any other storage device.

The first thing to do before inserting the disk into the drive is to set it to read only to avoid accidental alteration of the content. 3.5" floppy disks have a mode, similar to the one found on modern devices such as SD cards, and which can be activated with a physical mechanism located in a corner of the medium: by moving the appropriate slider to the "open hole" position, the disk is in read-only mode.

When you insert the disk into the drive, the operating system should activate the same process that comes into play when you insert, for example, a USB stick. If this does not happen or you are prompted to format the disk, something is wrong: it is possible that the disk is damaged or that it is not actually in PC / MS-DOS format. Conversely, if the operation was successful, we will be able to open a file manager window and explore the contents of the floppy. At this point, we can copy the data to the PC in a new folder created specifically for this purpose.

Taking a look at the contents, it should be easy to identify among the documents in the folder a main executable file (the extension should be .EXE, or .BAT, or .COM): take note of its name because it will be useful later.

Before adding content to the IA portal, it is a good idea to check that the electronic publication is “self-consistent”, that is, that it includes both the data and some software (there were dozens of them), designed to manage the hyperlink between the various “pages” of the magazine and to provide the user with a point-and-click interface for use, and that therefore the executables contained in the diskette do not require installation and do not presuppose the presence of other programs. At this stage you can also confirm

that the executable identified from its name actually corresponds to the one that starts the fanzine management interface.

To do this, you can use a PC/MS-DOS emulator directly on your PC. The program we recommend using is called dosbox-x, a free tool that can be found for free and is based on the dosbox program, used on IA. To familiarize yourself with this tool, we refer you to the project documentation, very complete and well written, and to the numerous “survival” guides for the MS-DOS environment available online.

Once you have made sure that everything is working correctly, you can proceed to upload the data. Uploading this data to IA follows the same procedure as any other archive, but to have the preview supported by the emulation services, you need to follow some precautions and specify three more parameters in the metadata editor.

First of all, it is necessary to “package” the contents of the diskette, including all the documents present in the archive/folder in a single ZIP file. This will be the main content to upload to archive.org, together with any other documents such as scans of the cover graphics or other.

As usual in the publication procedure, immediately

after uploading the files, you will have to enter the metadata, as described above. The metadata that interest us to activate the emulated previews are at the bottom of the form. In the *More options / Add additional metadata...* section we will have to add three parameters: *Emulator*, *Emulator_ext* and *Emulator_start*.

By specifying this information in the field/value format, we will specify respectively the name of the emulator that we want to activate, the format of the file that contains data and programs to be “fed” to the emulator and the name of the program to be executed immediately after starting the emulator. In the case of the example we are dealing with, the three field/value pairs will be (fig. 5):

Emulator dosbox ← the emulation service to be activated

Emulator_ext ZIP ← the format in which we have “packaged” the data

Emulator_start ADE4.EXE ← the name of the main executable to be started

At the end of the publication procedure, and after having waited a little while as in the cases seen previously, we should be able to use the programs in emulated mode on the archive web page.

In my experience, it may happen that – probably due to some small bug in the portal, which could be solved by the time you read this guide – the three additional

parameters mentioned above are not received by the system. To solve this potential problem, it will be sufficient to open the metadata editor after publication and add the field/value pairs again.

Click on any field below to edit it

Page Title *	adenoidi 11	✓
Page URL *	https://archive.org/details/ adenoidi-n11	✓
Description *	Add a description of the item page	✓
Subject Tags *	Add keywords, separated by commas	✓
Creator	Creator of the content	✓
Date	Date work was created/published	✓
Collection *	- - - Pick a collection - - -	✓
Test Item	No	✓
Language	Language of the work	✓
License	No license selected	✓
More Options Add additional metadata...		
emulator	dosbox	(remove)
emulator_ext	ZIP	(remove)
emulator_start	ADE4.EXE	(remove)

Fig. 5 - Example of metadata to enter to use software emulators

If something goes wrong in the operation, after a while your item will automatically end up in a very particular collection, Non-Working Software Contributions, a sort of limbo where software uploaded to IA but not working are temporarily deposited. If the person who uploaded them does not find a solution to the problem, after a few these non-working items are deleted.

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Internet Archive (www.archive.org) is a non-profit, free and open digital library, founded in 1996 with the aim of providing universal access to knowledge. On the platform, you can freely access millions of books, magazines, films, audio, software, and even web pages, including those of sites that no longer exist. Anyone, from major institutions to individual users, can create their own digital library on Internet Archive, sharing their digitized collections with a growing international audience. This guide provides a simple but detailed explanation of how to create your own collection on the world's most forward-thinking digital library and manage it effectively through careful use of metadata. In the appendix, an easy guide to using software emulators on Internet Archive.

English edition with a guide to using software emulators in the appendix.

